

ENVIRONMENTAL RESEARCH
LETTERS

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OPEN ACCESS

RECEIVED
12 April 2024REVISED
25 July 2024ACCEPTED FOR PUBLICATION
8 August 2024PUBLISHED
22 August 2024

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The transition to humid extreme cold events in winter during the
past six decades over ChinaYongli He^{1,2,*} , Yangrui Peng¹, Zhanbo Wang¹, Shanshan Wang¹ and Xiaodan Guan^{1,2} ¹ Key Laboratory for Semi-Arid Climate Change of the Ministry of Education, School of Atmospheric Sciences, Lanzhou University, Lanzhou, Gansu, People's Republic of China² Collaborative Innovation Center for Western Ecological Safety, Lanzhou, People's Republic of China

* Author to whom any correspondence should be addressed.

E-mail: heyongli@lzu.edu.cn**Keywords:** extreme cold events, humidity, wet-bulb temperature, China, humidity-dominatedSupplementary material for this article is available [online](#)

Abstract

Winter extreme cold events (ECEs) pose significant economic and health risks, particularly when cold is combined with humidity. Although global warming has led to a decline in overall ECE occurrences, the changes in humid ECEs remain uncertain. This study uses dry and wet-bulb temperatures to categorize dry and humid ECEs across China and analyzes the proportion of humid ECEs, and the underlying factors influencing this phenomenon. In contrast to the decreasing trend of winter relative humidity (RH), our findings show an increase in RH during ECEs. This observation suggests a shift toward more humid conditions during these events. Further analysis shows an increasing percentage of humid ECEs, especially in Xinjiang, where the occurrence of dry ECEs is decreasing. Classifying ECEs into temperature- or humidity-dominated types reveals a significant increase in humidity-dominated humid ECEs, which is driven mainly by specific humidity changes, not temperature changes. These findings highlight the urgent need for adaptation strategies for more humid ECEs in a warming climate.

1. Introduction

Extreme cold events (ECEs) are significant winter hazards marked by sharp temperature drops, frost, and snow that adversely affect agriculture, transportation, energy consumption, and health (Orlowsky *et al* 2011, Wi *et al* 2011, Yeh *et al* 2018, Herring *et al* 2021, IPCC 2021). The variability of ECEs complicates their responses to global warming (Wallace *et al* 2014). Despite a general trend toward fewer ECEs, evidence indicates an increase in severe ECE occurrence at mid-latitudes in the current century (Ma *et al* 2018, Yang *et al* 2020, Gao *et al* 2021, He *et al* 2023b). For instance, Xinjiang witnessed a record-breaking 24 h temperature drop (approximately 57 °C) in February 2024, highlighting the unpredictable nature of ECEs in a warming climate. This unpredictability underscores the need for continued exploration into how ECEs evolve with global warming.

Growing concerns about extreme temperature events, especially those influenced by water vapor,

have sparked significant research interest (Luo *et al* 2022, Li *et al* 2023, Zhang *et al* 2023c, Zhang *et al* 2023b). Unlike dry cold spells, cold and humid conditions pose greater health risks because high relative humidity (RH) can exacerbate the body's response to cold stress (Wang *et al* 2019, Yu *et al* 2021, Luo *et al* 2022, Zhang *et al* 2023a). High RH not only amplifies cold stress but also worsens discomfort during cold spells, highlighting the dual threat of humidity (Zhang *et al* 2023c). Thus, ECEs present a two-fold peril: they involve both chilling temperature and humidity-related challenges. Specifically, cold waves with high humidity can trigger severe cooling and precipitation events, mirroring the health hazards of humid ECEs (Zhang *et al* 2011, Qian *et al* 2014, Zheng *et al* 2022, Li *et al* 2023, Zhang *et al* 2023a). In this context, the wet-bulb temperature (TW), which captures both air temperature and humidity, becomes crucial for assessing humid ECEs (Stull 2011, Freychet *et al* 2020). Considering that the TW can reflect the influence of water vapor more accurately, it serves as an

effective indicator for identifying the characteristics of humid ECEs.

Although the effect of water vapor amplification is tangible, a unified approach for defining and identifying humid ECEs remains elusive. Current methodologies use various empirical indices, such as the heat stress index, apparent temperature, and wet-bulb globe temperature, to evaluate the impact of climate change on extreme temperature phenomena (Xu *et al* 2021, Li *et al* 2020, Luo *et al* 2022, Zhang *et al* 2023b). While informative, these indices are complex and pose measurement challenges. In contrast, *TW*, which is easily measurable with standard techniques, stands out as a pragmatic choice for climate research (e.g. Pal and Eltahir 2016, Raymond *et al* 2017). Its utility was demonstrated in studies like those by Luo *et al* (2022), who provided a method to categorize dry and humid heatwaves based on dry and wet-bulb temperature data. Nonetheless, the identification of humid ECEs and their dynamics under global warming remains a gap in our understanding. This gap highlights the importance of exploring the characteristics of humid ECEs and the significant role of moisture, pointing to a critical area for future research in the context of a changing climate.

Humid ECEs continue to occur in the context of global warming and are often accompanied by snow or rain, leading to substantial social impacts (Wang *et al* 2017). Freezing rain events in southern China during 2008 and 2024, characterized by both low temperatures and high humidity, caused extensive disruptions and economic losses. Additionally, a notable ECE in January 2008 brought substantial snowfall, impacting approximately 9 million people across 14 provinces (Cai *et al* 2019). These recent, frequent occurrences of cold-wet compound events raise the following question: will humid ECEs tend to increase compared to dry ECEs? Therefore, it becomes imperative to examine the changes in dry and humid ECEs in China, especially the influence of water vapor.

Our study categorizes ECEs into dry and humid types based on daily dry and wet-bulb temperatures, examines changes in the proportion of humid ECEs and identifies driving factors. The following questions will be addressed: (a) are humid ECEs becoming more frequent? (b) What is the effect of increasing water vapor on the changes in humid ECEs?

2. Materials and methods

2.1. Dataset

In this study, we used the fifth-generation European Center for Medium-Range Weather Forecasts atmospheric reanalysis (ERA5), which encompasses the multiple variables including surface air temperature (SAT), 2 m dew point temperature (d2m), and surface pressure (sp), over the period from 1959 to 2020. These variables have a spatial resolution of 0.25° and an hourly temporal resolution

(Hersbach *et al* 2020). The daily *TW* is calculated from the daily mean SAT, dew point temperature, and surface pressure, following methodologies outlined by the National Center for Atmospheric Research Command Language (www.ncl.ucar.edu/Document/Functions/Built-in/wetbulb.shtml). To corroborate our findings, the JRA-55 reanalysis dataset was also employed. Developed by the Japan Meteorological Agency, JRA-55 is recognized for the second Japanese global atmospheric reanalysis project, incorporating variables such as temperature, dew point temperature, and RH (Kobayashi *et al* 2015). It contains 37 pressure levels from 1 to 1000 hPa, including 16 layers below 500 hPa, across a grid of 288×145 points with a $1.25^\circ \times 1.25^\circ$ horizontal resolution and a 6 h temporal resolution. For analytical consistency, we adjusted the spatial resolution of JRA-55 to match that of the ERA5 dataset at $0.25^\circ \times 0.25^\circ$ through interpolation. We also used an observational dataset to verify the reliability of the results. The RH data from the gridded daily mean datasets CN05.1, developed by Wu and Gao (2013), are employed in this study, with a spatial resolution of $0.25^\circ \times 0.25^\circ$.

2.2. Definition of dry and humid ECEs

Following the approach detailed by Luo *et al* (2022), our study categorizes ECEs in China into two distinct categories: dry ECEs and humid ECEs. Dry ECEs are defined as conditions where $TW > TW_{10}$ and $T < T_{10}$ for at least three consecutive days, while humid ECEs are defined as conditions where $T > T_{10}$ and $TW < TW_{10}$ for at least three consecutive days. Here, T and TW represent the daily dry-bulb and wet-bulb temperatures, respectively. T_{10} and TW_{10} refer to the 10th percentiles of daily dry-bulb and wet-bulb temperatures, respectively. For each calendar day, the 10th percentiles for T and TW were calculated within a 3 day half-window centered on the day, spanning the period from 1959 to 2020. This was further smoothed by applying a 31 day moving average to eliminate high-frequency noise (He *et al* 2023a, Wang *et al* 2023, Peng *et al* 2024). At each grid point, we obtained the time series of T and TW . We then filtered out the time periods that satisfied the following conditions: $TW > TW_{10}$ and $T < T_{10}$ for at least three consecutive days were defined as dry ECEs, and $T > T_{10}$ and $TW < TW_{10}$ for at least three consecutive days were defined as humid ECEs. An occurrence must persist for a minimum of three consecutive days to be classified as a humid or dry ECE, thereby filtering out short events caused by local temperature fluctuations. This classification scheme allows us to isolate and analyze ‘pure’ dry and humid ECEs, providing clearer insight into the distinct dynamics and implications of each event type. The hybrid ECEs, where $T < T_{10}$ and $TW < TW_{10}$, are excluded from this analysis, as they do not fall neatly into the ‘pure’ dry or humid ECEs and represent a smaller proportion of

regionally specific subset, which is primarily observed in Xinjiang and Northeast China (figures S1–S3).

2.3. Excursions of T and q to dry and humid ECEs

We employed a standard-anomaly ratio (SAR) method to determine the specific contributions of temperature (T) and specific humidity (q) to the occurrence of dry and humid ECEs across China. This approach, aligned with methodologies used in previous studies (Raymond *et al* 2017, Wang *et al* 2019), focuses on deviations beyond extreme thresholds rather than average conditions. The SAR is defined by the following relationship:

$$\text{SAR} = \begin{cases} |R_1/R_2| & R_1 < 0 \ \& \ R_2 > 0 \\ 0 & R_1 \geq 0 \ \& \ R_2 > 0 \\ \text{maximum} & R_1 < 0 \ \& \ R_2 \leq 0 \\ 1 & R_1 \geq 0 \ \& \ R_2 \leq 0 \end{cases} \quad \begin{matrix} (1) \\ (2) \\ (3) \\ (4) \end{matrix}$$

where $R_1 = [(T - T_c)/T_s]$ and $R_2 = [(q - q_c)/q_s]$, represent the normalized deviations of T and q from their respective extreme thresholds (T_c and q_c) normalized by their standard deviations (T_s and q_s) within a 7-day centered moving window during 1959–2020. Here, T_c and q_c denote the daily thresholds for extreme T and q , q_c are the 90th percentile values of specific humidity on that calendar day and T_c is the 10th percentile value of temperature on that calendar day. T_s and q_s are the corresponding standard deviations. Since the temperature in winter can be below zero, we use the absolute value for equation (1). In equation (3), ‘maximum’ refers to the maximum value of the SAR in equation (1). Four scenarios are outlined based on the SAR value, enabling a nuanced analysis of the impact of T and q : both T and q are extreme (Case 1); q is extreme but T is not (Case 2); T is extreme but q is not (Case 3); and neither T nor q is extreme (Case 4). An SAR greater than 1 signifies that temperature variations predominantly drive the change, whereas an SAR less than 1 indicates a greater influence from specific humidity. This distinction allows for the classification of each ECE as either T -dominated or q -dominated, providing a comprehensive framework to assess the relative contributions of T and q to the dynamics of dry and humid ECEs.

2.4. Statistical methods

For the regional analysis, we assessed trends at each grid point, employing a 5%–95% range for uncertainty unless specified otherwise. Composite analyses were conducted to reveal the characteristics of RH and temperature anomalies for both dry and humid ECEs. Daily anomalies were calculated by subtracting the climatological cycle from the original daily values. The long-term trends of daily anomalies are then removed by using simple linear regression to eliminate the possible effects of climate change on atmospheric circulation. Here the climatological cycle is

obtained by averaging the calendar day over the reference period of 1959–2020. Following the methodology of Luo *et al* (2022), the composite mean for an individual ECE is obtained by averaging all the days involved in that specific event. The composite map for all ECEs was then constructed by averaging the composite means of all individual ECEs. Despite the variation in the timing of ECEs among different grids, the anomalies are calculated by subtracting the corresponding climatological values for each grid and calendar day. Therefore, this approach ensures that the temporal differences in ECEs do not affect the composite anomaly.

3. Results

3.1. Increased RH during ECEs

Against the background of enhanced warming in the winter season, especially in semi-arid regions, as noted by Huang *et al* (2012), a decrease in RH has been observed, as shown in figures 1(a) and S4(b). However, recent findings indicate that although the frequency of ECEs is decreasing, their intensity remains unchanged (Cadiou *et al* 2024). He *et al* (2023a) also reported an increase in strong ECEs and a decrease in weak ECEs. These findings drove the investigation into whether the RH during total ECEs showed a consistent decreasing trend. Figure 1(a) shows the time series of RH during the winter season and of ECEs during 1959–2020 across China. The total ECEs here refer to the sum of dry and humid ECEs classified according to the definition in section 2.2. We focus exclusively on ECEs in winter. There was a significant decrease in RH across winter, with a rate of -0.43% decade $^{-1}$, in contrast to an increase in RH during total ECEs of 0.25% decade $^{-1}$ (figure 1(a)). Figures 1(b)–(d) provides a detailed comparison of regional differences in RH trends during dry ECEs, humid ECEs and total ECEs. All trends were rigorously tested for statistical significance, confirming their validity at the 95% confidence level. Notably, there is a significant trend of increasing RH during total ECEs in China, particularly in the northern region (figure 1(d)). However, data gaps in regions such as Northeast China, Qinghai, and Xinjiang, resulting from the infrequent occurrence of total ECEs (refer to figures S2 and S3), restrict our ability to calculate trends. At the same time, the RH during dry ECEs shows a significant decreasing trend (figure 1(b)) across the entire region, while the RH during humid ECEs exhibit a significant increasing trend (figure 1(c)). Therefore, the non-significant increasing trend observed in the total ECEs in figure 1 is driven primarily by humid ECEs rather than dry ECEs. We also investigate the trend of RH during ECEs defined by temperature (figure S4(a)) and during winter (figure 4(b)). Our results indicate that the trend of RH during ECEs defined by T generally shows a decreasing trend across most regions, with

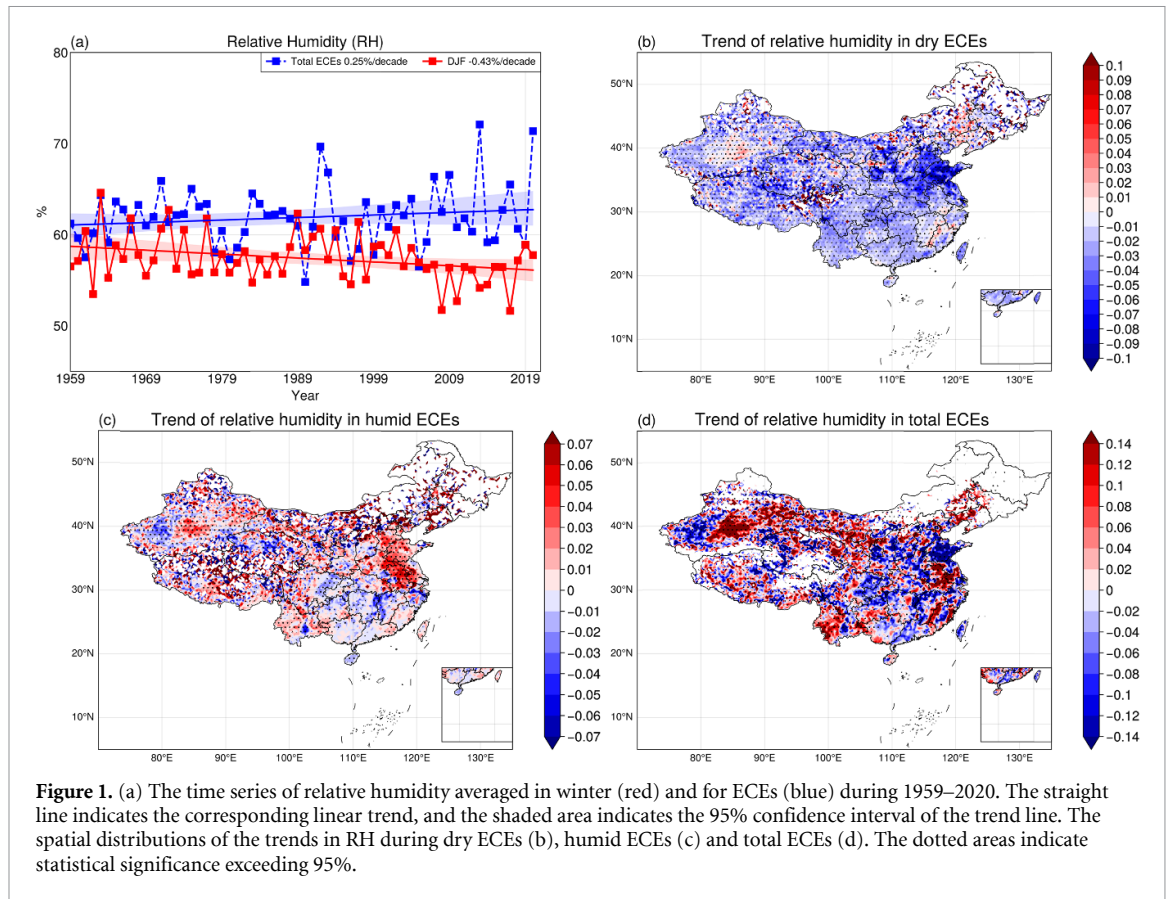


Figure 1. (a) The time series of relative humidity averaged in winter (red) and for ECEs (blue) during 1959–2020. The straight line indicates the corresponding linear trend, and the shaded area indicates the 95% confidence interval of the trend line. The spatial distributions of the trends in RH during dry ECEs (b), humid ECEs (c) and total ECEs (d). The dotted areas indicate statistical significance exceeding 95%.

the exception of specific areas. The winter-averaged RH also generally shows a decreasing trend, except in part of Xinjiang (figure S4(b)). Additionally, evidence from the JRA-55 and CN05.1 datasets supports the observed increasing RH trend during total ECEs, strengthening these findings (figures S5 and S6).

3.2. The transition to humid ECEs

As detailed in section 2.2, this study classifies ECEs into dry and humid ECEs based on the conditions of T and TW . A crucial question arises: does an increase in RH lead to the transition of ECEs toward humid ECEs? Before exploring their transitions, we first examined the differences between dry and humid ECEs. Figure S7 shows the spatial patterns of composite anomalies in the near-surface air temperature (T) and RH for dry and humid ECEs in China from 1959 to 2020. During dry ECEs, southern China and northern Inner Mongolia exhibited relatively mild negative temperature anomalies (figure S7(b)), which were significantly lower than those observed during humid ECEs (figure S7(a)). The RH during dry ECEs shows a notable negative anomaly in southern China (figure S7(d)), while certain areas in the northeast and Tibetan Plateau show a positive anomaly. In contrast, during humid ECEs, the RH exhibits positive anomalies, especially in Northwest China, where it is 40% higher than the climate average state (figure S7(c)).

These contrasting spatial patterns in the RH anomalies highlight the distinct impacts of temperature and humidity on dry and humid ECEs, confirming the effectiveness of the proposed classification scheme.

We introduce a quantitative measure, the ‘ratio index’, which is defined as the ratio of the frequency of humid ECEs to the total number of dry and humid ECEs. An increasing trend in this ratio indicates a shift from dry to humid ECEs, implying a future scenario where humid ECEs may become more prevalent. Temporal analysis, illustrated in figure 2(a), reveals discernible declining trends across all categories of ECEs. The humid ECEs exhibited the slowest decline rate ($-0.13 \text{ decade}^{-1}$), in contrast to the more pronounced reduction in the total ECEs ($-0.43 \text{ decade}^{-1}$). Notably, figure 2(b) highlights a rising trend in the ratio of humid ECEs to total ECEs at a rate of 2.43%/degrees. However, this trend is not statistically significant, prompting a detailed spatial examination of the trend map of this ratio. Spatially, with the exception of Jiangxi, Hunan, Hubei and Guangxi provinces, a significant increasing trend in this ratio was observed across most of China. Inner Mongolia emerges as the region exhibiting the most substantial increase, with a growth rate of 2.4 yr^{-1} . These findings suggest a nuanced shift to humid ECEs, where despite an overall decline in event frequency, the proportion of humid ECEs relative to the

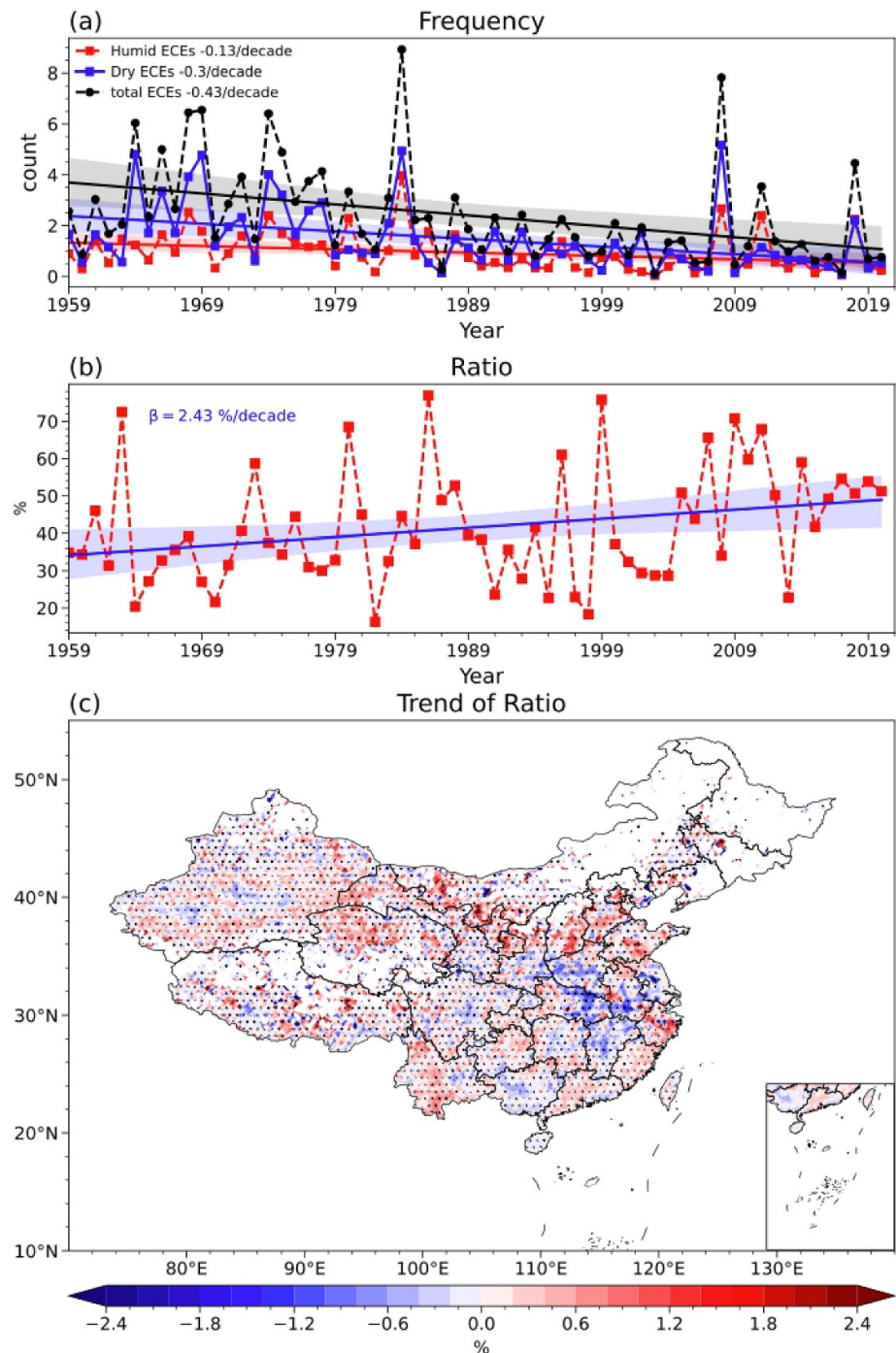
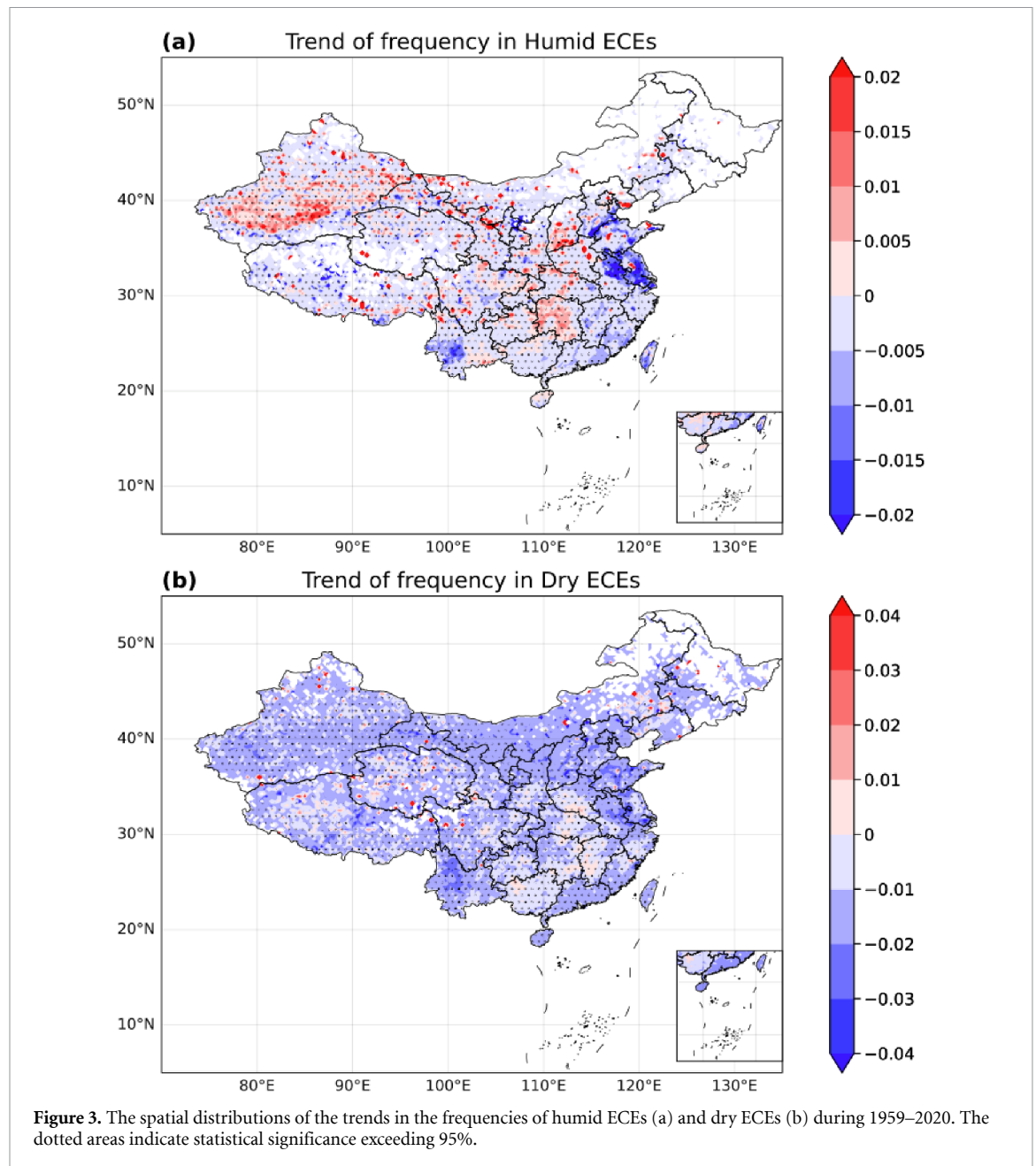


Figure 2. (a) Time series of the frequencies of dry ECEs, humid ECEs and total ECEs in winter during 1959–2020. (b) The time series of the ratio of humid ECEs to total ECEs during 1959–2020. The straight line indicates the corresponding linear trend, and the shaded area indicates the 95% confidence interval of the trend line. (c) The spatial distribution of the trend in the ratio of humid ECEs to total ECEs. The dotted areas indicate statistical significance exceeding 95%.

total is expanding. This trend of the ratio underscores a gradual transition toward a future climate regime characterized by a higher prevalence of humid ECEs. We also recalculated the distribution of the ratio of the frequency of humid ECEs to the total frequency of dry, humid and hybrid ECEs, as shown in figure S8. These results are consistent with those obtained previously.

3.3. The predominant driver behind the transition to humid ECEs

Figure 3 displays the trend map for the frequency of dry and humid ECEs. The frequency of humid ECEs significantly increased in Northwest China and Hunan Province, with the largest increase observed in northern Xinjiang at $0.02 \text{ count yr}^{-1}$. In western Jiangsu and Anhui Provinces, the frequency of humid

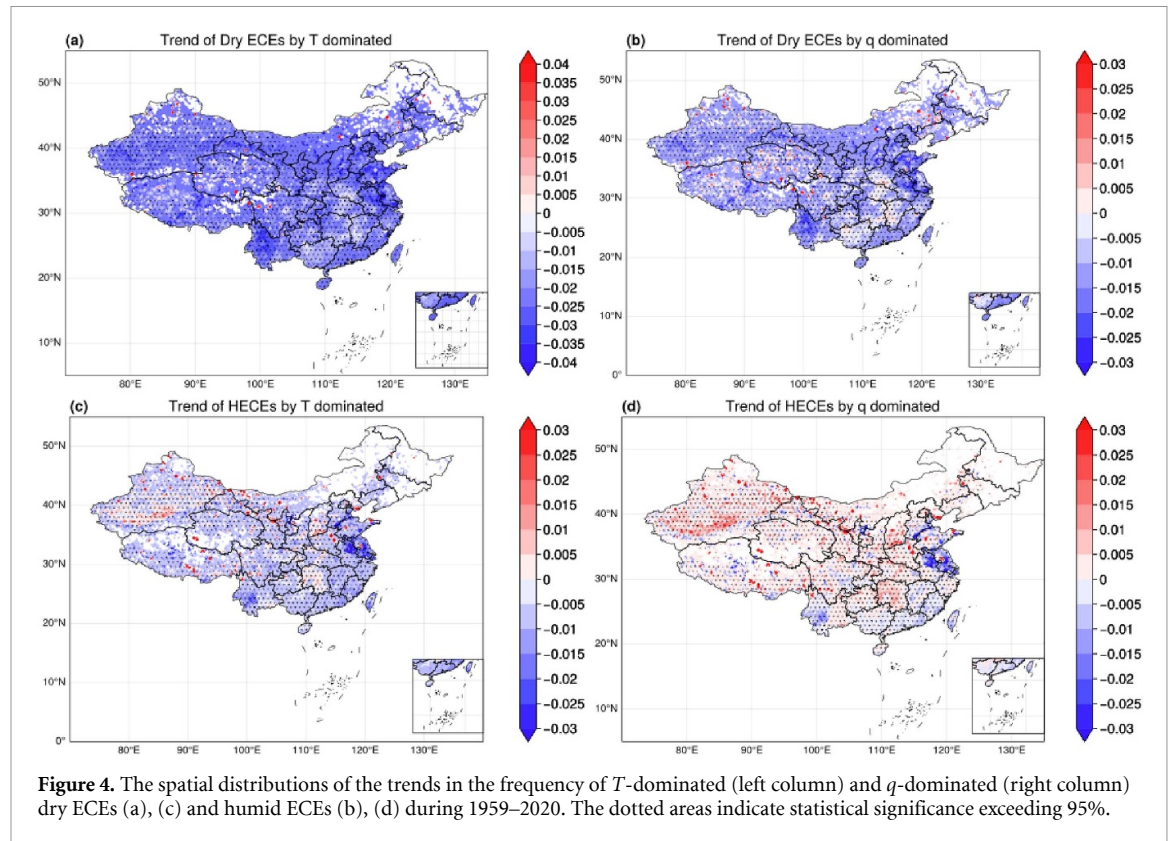


ECEs showed the most significant decrease, with a trend of -0.02 yr^{-1} (figure 3(a)). In contrast to that of humid ECEs, the frequency of dry ECEs exhibited a consistent significant decrease across China (figure 3(b)). The contrasting trend in the frequency of dry and humid ECEs indicates that the proportion of humid ECEs is increasing. To further investigate the impact of temperature and humidity on the proportion increase, we analyzed the frequency of dry and humid ECEs dominated by T and q .

An ECE can be classified as temperature-dominated or humidity-dominated using a SAR method as detailed in section 2.3. Figures S9 and S10 show the spatial distributions of the proportions of the four conditions under which dry and humid ECEs occur. Across China, dry ECEs in Case 3 are much more frequent than those in Cases 1 and

2 (figure S9(c)), which suggests that dry ECEs are more strongly affected by temperature than humidity. Extreme T and extreme q usually co-occur during humid ECEs (Case 1), especially in northern China (figure S10(a)), and Case 2 with extreme q but non-extreme T , occurs more often (figure S10(b)), suggesting that humid ECEs are primarily favored by atmospheric conditions that are either very humid and very cold or very humid but not very cold. More details are described in text S1 in the supporting supplementary information.

Here, we disentangle the contributions of temperature (T) and specific humidity (q) to dry and humid ECEs in China. Figure 4 shows the trends in the frequency of dry and humid ECEs dominated by temperature and humidity, respectively. Both the T -dominated and q -dominated dry ECEs exhibited



decreasing trends. However, the decreasing trend of the *q*-dominated dry ECEs was relatively weaker than that of the *T*-dominated dry ECEs (figures 4(a) and (b)). The *T*-dominated humid ECEs increased significantly in northern Xinjiang and Hunan Provinces, but decreased elsewhere, with the most obvious decreases occurring in Jiangsu and western Anhui Provinces (figure 4(c)). The *q*-dominated humid ECEs showed an increasing trend across most regions of China, with the most significant increase observed in Xinjiang, with a trend of $0.03 \text{ count yr}^{-1}$. However, a decreasing trend persists in some southeast coastal provinces, most notably in western Jiangsu and Anhui Provinces, with a trend of -0.03 yr^{-1} (figure 4(d)). These findings suggest an increasing prevalence of *q*-dominated humid ECEs, contributing to a higher proportion of humid ECEs in the overall frequency of ECEs.

4. Conclusions and discussion

This study identifies dry and humid ECEs in China during 1959–2020, based on dry and wet-bulb temperatures, and explores the transition toward more humid ECEs and their driving factors. We observe a notable increase in RH during ECEs, which contrasts with a significant decrease in the winter average RH. This phenomenon demonstrated that the ECEs are becoming more humid. This phenomenon can be attributed to enhanced winter warming, which raises the atmospheric water vapor capacity (Zhang *et al*

2023a). However, the faster rise in saturated water vapor pressure compared to the increase in specific humidity leads to reduced winter RH. In contrast, the temperature during ECEs has not shown a warming trend, with recent studies indicating that the intensity of ECEs remains stable (He *et al* 2023b, Cadiou and Yiou 2024). Therefore, the increase in specific humidity, along with stable temperature, contributes to the increasing RH during ECEs.

To explore the influence of increasing RH on the dry and humid ECEs, we examined the ratio of humid ECEs to total ECEs. Our findings indicate an increasing trend in this proportion, suggesting a transition from dry to humid ECEs. Further analysis of the frequency trends reveals a marked decline in dry ECEs and a significant increase in humid ECEs, particularly in northern China. This shift underscores the transition toward more humid ECEs, significantly attributing this to increased atmospheric water vapor. The increase in water vapor notably impacts the long-term behavior of humid ECEs, offsetting the expected decline from global warming effects. We classify dry and humid ECEs into *T*-dominated and *q*-dominated type using the SAR method and further study their contributions to the increasing proportion of humid ECEs. Our analysis reveals a decrease in both *T*-dominated and *q*-dominated dry ECEs, whereas *q*-dominated humid ECEs significantly increase. These trends lead to an increasing ratio index of humid ECEs, emphasizing the critical role of specific humidity in this transition. To verify the reliability of our

results, we recalculated the frequency of dry and humid ECEs based on the number of days and further verified the results in figures 2–4. Although the trend magnitudes differ, all the trend distributions are consistent with the previous results (figures S11–S13).

The mechanism behind the shift to humid ECEs is partly explained by the dual effects of global warming on temperature and water vapor. On the one hand, global warming directly decreases the frequency of ECEs by increasing temperature. On the other hand, global warming increases water vapor content, which could increase the probability of humid ECEs due to the amplification effect of water vapor as reported in recent studies (Zhang *et al* 2023a, Peng *et al* 2024). As a result, the rate of decrease for humid ECEs is slower, leading to an increased ratio of humid ECEs within the total ECEs. This increasing trend is more pronounced in the northern China, which is consistent with the observed increase in RH during humid ECEs in these regions.

This study provides a novel perspective on the changing nature of ECEs in a warming climate, highlighting a significant shift toward more humid ECEs. This transition, evidenced by increasing q -dominated humid ECEs, signifies a critical deviation from dry ECEs. While existing literature has explored individual instances of compound events with high humidity and low temperature, such as the cold-wet event in southern China in February 2022 and freezing rain incidents in 2008. The systematic analysis of the trend of humid ECEs in this study aids in enhancing risk assessment and management strategies for such compound extreme events.

Despite these advances, the physical mechanisms driving this transition to humid ECEs remain inadequately understood and require further research. The complex interactions among global warming, atmospheric circulation patterns and the specific processes contributing to the shift of humid ECEs are not yet full unraveled. For example, Li *et al* (2023) found that the severe compound cold-wet event in southern China in February 2022 is jointly regulated by La Nina-like SST conditions in the tropical Pacific and warm SST anomalies in the North Atlantic. Therefore, a thorough analysis of the associated atmospheric and oceanic dynamics is essential for fully understanding the trends of humid ECEs and their implications for future climate scenarios. Advancing this line of inquiry is crucial for enhancing our understanding and improving the accuracy of predictive models for future extreme weather trends.

Data availability statements

The data supporting the findings are freely available. The ERA5 dataset is available from the website (<https://cds.climate.copernicus.eu/cdsapp#!/dataset/reanalysis-era5-single-levels>). The JRA-55 dataset is

available from the website (<https://doi.org/10.5065/D60G3H5B>).

Acknowledgments

This work was jointly supported by the National Science Foundation of China (42375021) and Central Guidance Fund for Local Science and Technology Development Projects in Gansu (No. 24ZYQA031). This work was also supported by the Supercomputing Center of Lanzhou University. This work was supported by the National Key Scientific and Technological Infrastructure project ‘Earth System Science Numerical Simulator Facility’ (EarthLab).

ORCID iDs

Yongli He  <https://orcid.org/0000-0002-0183-378X>
Xiaodan Guan  <https://orcid.org/0000-0003-3716-4503>

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